INCH-POUND

MIL-W-43685B(GL)
20 September 1989
SUPERSEDING
MIL-W-43685A(GL)
23 June 1975

MILITARY SPECIFICATION

WEBBING AND TAPE, TEXTILE, ARAMID FIBER

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers webbing and tape made from aramid fiber.

1.2 Classification. The webbing and tape shall be of the following types as specified (see 6.2).

Type I - Double woven webbing
Type II - Single woven tape

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATION

MILITARY

MIL-P-43334 - Packing of Textile Webbing and Tape

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Standard sample. The solution-dyed webbing and tape shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

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3.3 Materials

3.3.1 Yarn. The yarn for the warp and filling shall be made from a bright continuous filament aramid fiber (see 6.6). The yarn may be heat set (see 6.5) to meet the shrinkage requirement specified in 3.8. The yarn denier and yarn ply shall be as specified in table I. When ply yarns are specified, the final ply of the yarn shall have a minimum of 2-1/2 turns per inch (TPI). The number of single yarns specified in table I shall be twisted together (plied) in one operation. Testing shall be as specified in 4.4.1.2.

3.4 Color. The color of the webbing and tape shall be Olive Green 106 to match the standard sample when tested as specified in 4.4.3. The color shall be obtained by the use of melt spun solution-dyed fibers.

3.4.1 Matching. The color of the finished webbing and tape shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.4.2 Colorfastness. The finished webbing and tape shall show fastness to light and laundering equal to or better than the standard sample or equal to or better than a rating of "fair" for fastness to light and "good" for fastness to laundering. Testing shall be as specified in 4.4.3.

3.5 Physical requirements. The construction and physical requirements of the finished webbing and tape shall be as specified in table I.

TABEL I. Physical requirements

	Type I	Type II
Width (inch)	5/8 -1/32 +1/16	1 ± 1/32
Thickness (inch)	0.040 min. -0.055 max.	0.015 min. -0.025 max.
Weight oz./lin. yd. (max)	0.45	0.30
Breaking strength, lb. (min)	650	525
Ends in warp (min)		
Face and back	158	276
Binder	9	-
Stuffer	40	-
Filling yarns per inch (min)	48	30

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TABEL I. Physical requirements (cont'd)

	Type I	Type II
Yarn denier (nominal)		
Face and back	2 ply 200 denier	200 denier
Binder	2 ply 200 denier	-
Stuffer	2 ply 200 denier	-
Filling	4 ply 200 denier	2 ply 200 denier

3.6 Weave. The weave for the webbing and tape shall be as specified in 3.6.1 and 3.6.2. The filling yarn for both types shall traverse the full width of the webbing or tape with one filling yarn per shed.

3.6.1 Type I. The type I webbing shall be a double fabric consisting of face and back warps with two ends weaving as one and a stuffer warp, all bound together by a binder warp with the ends weaving singly. The stuffer ends shall weave between the face and back warps and shall not show on either the face or the back. The face warp shall weave plain weave with the picks that show on the face. The back warp shall weave plain weave with the picks that show on the back. The binder warp shall weave plain weave throughout.

3.6.2 Type II. The type II tape shall be plain weave, except that the warp ends shall weave two ends as one.

3.7 Finish. The webbing and tape shall be scoured. Heat setting (see 6.5) is required in either the yarn or in the webbing and tape in order to meet the shrinkage requirement specified in 3.8.

3.8 Shrinkage. The shrinkage of the finished webbing and tape shall be no more than 5.0 percent when tested as specified in 4.4.3.

3.9 Length and put-up. For Government procurement only, unless otherwise specified (see 6.2), the webbing and tape shall be furnished in lengths of 90 to 110 yards. No roll shall contain more than three pieces and no piece shall be less than 3 yards in length.

3.10 Identification tickets. Each roll of webbing and tape shall have an identification ticket attached to the roll in accordance with MIL-P-43334.

3.11 Fiber identification. Each roll of webbing and tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished webbing and tape shall conform to the quality of product established by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 through 4.4.2.3 and tested as specified in 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the webbing and tape conforms to the requirements specified in 3.3.1 and table I.

4.4.1.2 Component testing. In addition to the quality assurance provisions of the subsidiary specifications, component materials listed in table II shall be tested for the specified characteristics in accordance with the referenced test methods of FED-STD-191. The lot size shall be expressed in pounds, and the sample unit shall be 500 yards of the nomex yarn. The lot shall be unacceptable if one or more units fail to meet any requirement specified. All test reports shall contain the individual values in expressing the final results. The sample size shall be as follows:

TABLE II. Component tests

Characteristic	Requirement paragraph	Test method
Material:		
Nomex identification	3.3.1	1/
Yarn:		
Twist	3.3.1	4054
Denier	3.5	4021 1/
Ply	3.5	Visual

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. The webbing and tape shall be examined on both sides for the defects listed in table III. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in linear yards. The sample unit shall be 1 linear yard of webbing or tape. The number of rolls from which the sample is to be selected shall be in accordance with table IV. The sample yardage shall be apportioned equally among the selected rolls. The inspection level shall be I (see 6.7).

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TABLE III. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Twist or distortion	Webbing or tape will not lay flat upon application of manual pressure due to twist or distortion.		201
Cut, hole, or tear	Any cut, hole or tear.	101	
Drop-ply	Clearly visible on more than two ends within same length and extending over 9 linear inches or more. 1/	102	
	Clearly visible on one or two ends within same length and extending over 9 linear inches or more. 1/		202
Edges	Frayed, slack or otherwise poorly constructed and exceeding 1/4 inch in length.	103	
Knots	More than one knot in any 9 linear inches.	104	
	More than one knot every 2 yards with untrimmed ends extending from surface of webbing.		203
Mispick, double pick	Two or more across the full width.	105	
Width	Beyond specified tolerances.		204

1/ Clearly visible at normal inspection distance (approximately 3 feet).

TABLE IV. Sample size

Lot size in yards	Sample size in rolls
1200 or less 1/	3
1201 up to and including 3200	5
3201 up to and including 10,000	8
10,001 up to and including 35,000	13
35,001 up to and including 150,000	20
150,001 and over	32

1/ If a lot contains fewer than 3 rolls, each roll in the lot shall be examined.

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4.4.2.2 Overall examination. The webbing and tape shall be examined for the defects listed below. Each defect listed shall be counted not more than once in each roll examined. The sample shall consist of the applicable number of rolls indicated in table IV (see 6.7).

Defects

Off shade, i.e., not within established tolerance

Uneven weaving throughout

Not labeled in accordance with Textile Fiber Products Identification Act

4.4.2.3 Length examination. Each roll of webbing and tape shall be examined for the defects listed below. The sample shall consist of the applicable number of rolls indicated in table IV. If the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets, the lot shall be rejected (see 6.7).

Defects

Gross length less than specified minimum length or more than specified maximum length

Gross length more than 2 yards less than gross length marked on roll tickets

Any piece less than 3 yards in length

Any roll containing more than three pieces

4.4.3 End item testing. The webbing and tape shall be tested for the characteristics listed in table V. Except where otherwise specified, the methods of testing specified in FED-STD-191 wherever applicable and as listed in table V shall be followed. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of 1 linear yard. The sample unit for testing shall be 10 yards. The lot shall be rejected if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE V. End item tests

Characteristics	Requirement paragraph	Test method	No. deter. per sample unit	Results reported as
Matching	3.4.1	9010	-	-
Colorfastness to:				
Light	3.4.2	5660 2/	-	-
Laundering	3.4.2	5610	-	-
Thickness	3.5	5030	-	-
Weight	3.5	5041	-	To nearest 0.01 ounce
Breaking strength	3.5	4108	-	-
Warp ends (full width):				
Face and back	3.5	5050	-	-
Binder (type I)	3.5	5050	-	-
Stuffer (type I)	3.5	5050	-	-
Filling yarns per inch	3.5	5050	-	-
Weave	3.6	Visual	1	Pass or fail
Finish:				
Scoured	3.7	1/	-	-
Heat set	3.7	1/	-	-
Shrinkage after five launderings	3.8	5556 3/	-	-

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Except that the contractor's submission shall be compared with the standard sample or rated after an exposure of 6 standard fading hours.

3/ Except that the length of the test specimen shall be 24 inches and the cotton laundering procedure shall be used. The stack temperature of the tumble drier shall be 180°F and the webbing or tape shall not be pressed.

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4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-P-43334.

4.4.5 Palletization examination. The examination shall be made in accordance with the quality assurance provisions of MIL-P-43334.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A, or Commercial as specified (see 6.2).

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, or Commercial. Webbing shall be packed in accordance with the applicable requirements of MIL-P-43334.

5.3 Palletization. When required, palletization shall be in accordance with the applicable requirements of MIL-P-43334.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with MIL-P-43334.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type I webbing is utilized as a slide fastener thong for Shirt, Man's, Army Aviation, Crewmember and as a waist adjustment strap for the trousers of Uniform, Army Aviation Crewmember. The type II tape is utilized as a retrieval strap in Coveralls, Armored Vehicle, Crewmember.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. When length of rolls is other than specified (see 3.9).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. When palletization is required (see 5.3).
h. Acceptance criteria required (see 6.7).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Heat set. An autoclave heat set procedure which has proven satisfactory is as follows:

5 minutes pre-vacuum
20 minutes 30 psi steam
5 minutes pre-vacuum
20 minutes 30 psi steam
5 minutes pre-vacuum
Exhaust to air

6.6 Fiber identification. The requirements of the webbing and tape are based on materials made from "NOMEX" aramid fiber manufactured by E. I. duPont deNemours and Company, Inc., Wilmington, DE 19800.

6.7 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

6.7.1 For yard-by-yard examination. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 0.65 for major defects and 4.0 for total (major and minor combined) defects is recommended.

6.7.2 For overall and length examination. The following acceptance numbers are recommended:

For sample size (rolls)	Acceptance number
Up to 13	0
20	1
32	2

The acceptance criteria as specified in the contract or purchase order shall be binding.

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6.8 Subject term (key word) listing.

Filament
Nomex
Synthetic
Uniforms

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - AV, MD
DLA - CT

Preparing activity:

Army - GL

(Project 8305-A283)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-43685B(GL)	Webbing and Tape, Textile, Aramid Fiber

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify): _____
3b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	7b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
7c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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